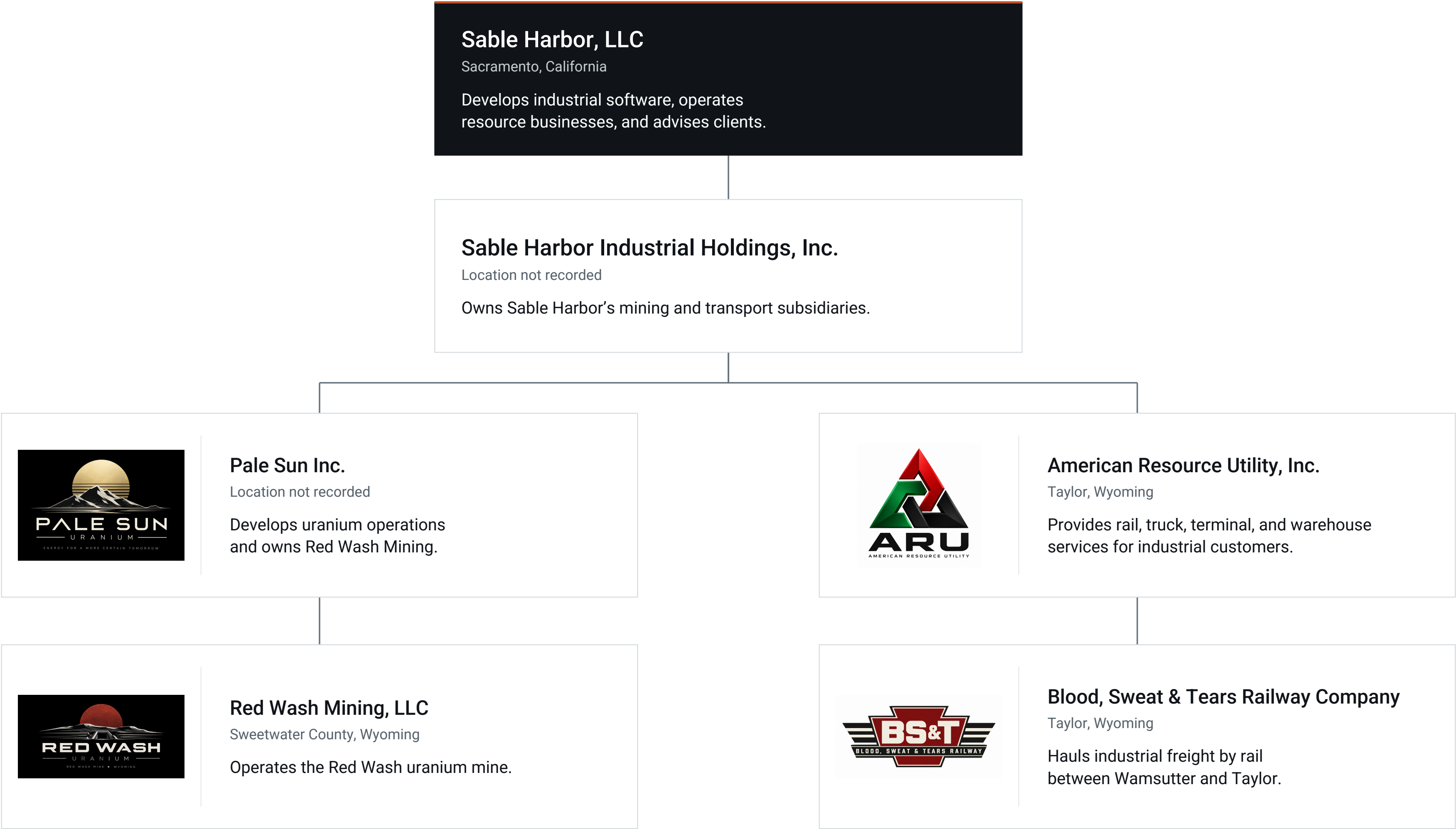


Business-line membership. Legal ownership is shown separately.



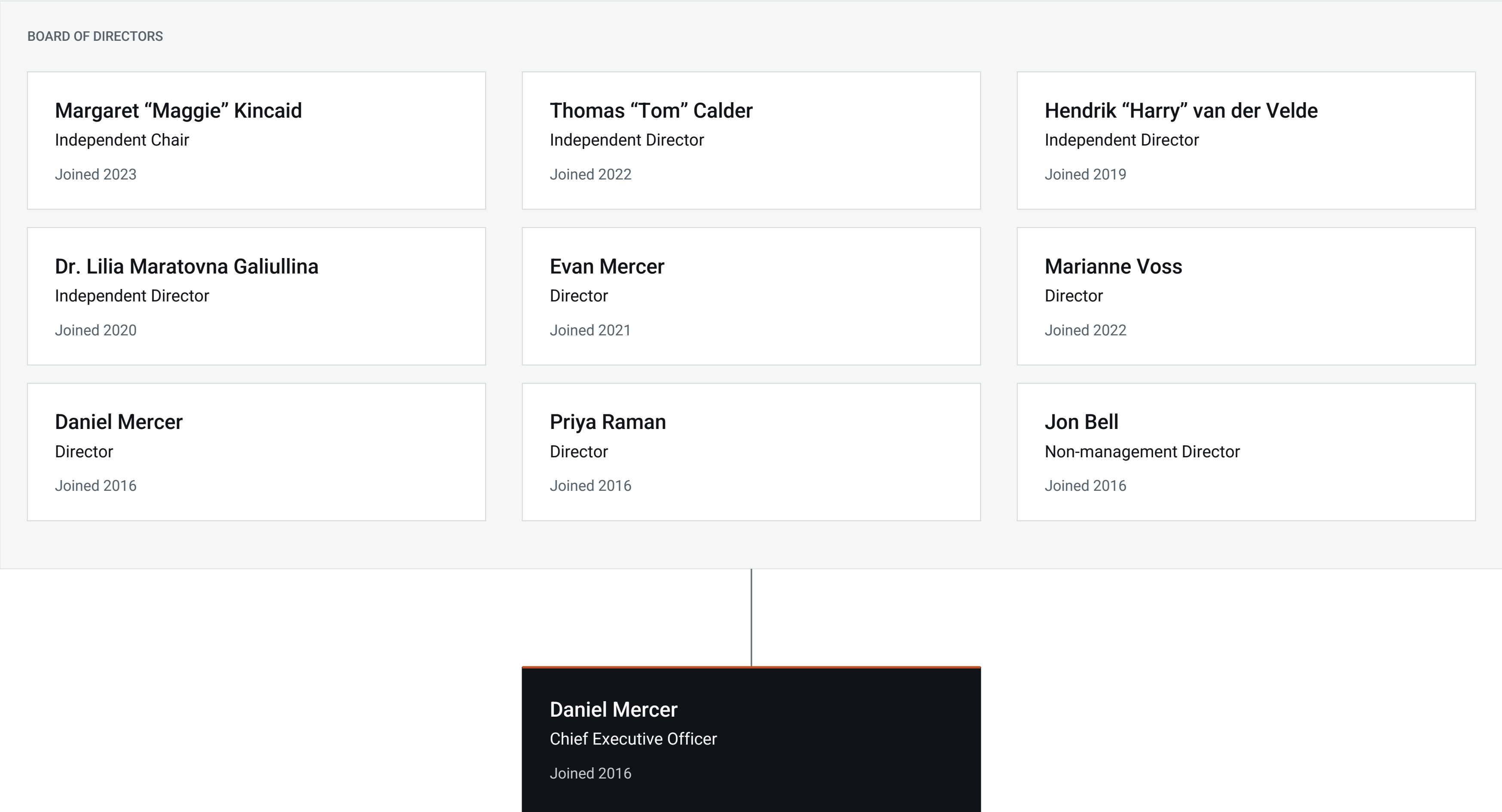
Industrial ownership

Each connector represents 100% equity ownership.



Board and Chief Executive

The Board collectively oversees the Chief Executive Officer. No individual director is shown as the CEO's separate manager.



Board cards use Board joining years. The CEO card uses company joining year.

Sable Harbor Advisory

Grouped membership; no reporting line or legal ownership is asserted.

Business



Sable Harbor Advisory

Location not recorded

Investigates business problems, builds client intelligence teams, and improves operations.

Practices

Intelligence Advisory

Location not recorded

Investigates business questions and assesses evidence for client decisions.

Practices

Intelligence Capability

Location not recorded

Builds clients’ intelligence teams, research processes, training, and supporting systems.

Practices

Operational Excellence

Location not recorded

Improves mining, logistics, maintenance, reclamation, and other client operations.

Shared delivery team

Advisory professional team

Location not recorded

Delivers client engagements using operators, analysts, engineers, and other specialists.

Independent verification

Value Office

Location not recorded

Verifies engagement baselines, measured results, and outcome-based fees.

Grouped membership; no reporting line or legal ownership is asserted.

Houses ARU's main rail yard, terminal, warehouse, trucking depot, and offices.

Transfers industrial bulk cargo between railcars, trucks, and storage.

Stores customer goods and handles rail and truck deliveries.

Dispatches trucks and provides vehicle parking, inspections, and maintenance.

ARU operating facilities

Grouped membership; no reporting line or legal ownership is asserted.

Taylor

Taylor ARU office

Taylor, Wyoming

Houses ARU administration, shared services, and customer support.

Rawlins

Rawlins satellite terminal

Rawlins, Wyoming

Handles industrial cargo delivered by truck.

Rawlins

Rawlins warehouse

Rawlins, Wyoming

Stores contracted customer inventory and picks orders.

Rawlins

Rawlins trucking satellite

Rawlins, Wyoming

Stages heavy-haul trucks and equipment.

ARU businesses and services

Grouped membership; no reporting line or legal ownership is asserted.

Operating company



American Resource Utility, Inc.
Taylor, Wyoming

Provides rail, truck, terminal, and warehouse services for industrial customers.

Railway subsidiary



Blood, Sweat & Tears Railway Company
Taylor, Wyoming

Hauls industrial freight by rail between Wamsutter and Taylor.

Services

Terminal operations
Taylor and Rawlins, Wyoming

Transfers bulk materials between trains, trucks, and storage.

Services

Trucking
Taylor and Rawlins, Wyoming

Hauls bulk materials, industrial supplies, and heavy equipment by road.

Services

Warehousing
Taylor and Rawlins, Wyoming

Stores customer inventory and picks and dispatches orders.

Services

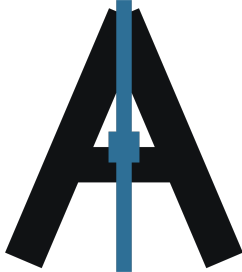
ARU corporate services
Taylor, Wyoming

Handles finance, personnel, customer service, and administration.

Atlas Meridian

Grouped membership; no reporting line or legal ownership is asserted.

Product



Atlas Meridian

Location not recorded

Software for investigating operating data, reviewing evidence, and managing client work.

Product capabilities

Product engineering

Location not recorded

Builds and maintains Atlas Meridian software.

Product capabilities

Product management and design

Location not recorded

Plans product features and designs how customers use Atlas Meridian.

Product capabilities

AI and agent evaluation

Location not recorded

Tests Atlas Meridian agents for accuracy, reliability, and permitted behavior.

Atlas Meridian

Grouped membership; no reporting line or legal ownership is asserted.

Product capabilities

Security, tenancy, and entitlements

Location not recorded

Secures the product and controls client access and data separation.

Product capabilities

Transfer and developer tooling

Location not recorded

Packages client workflows and provides tools to deploy and maintain them.

Product environments

Client Plane

Location not recorded

Holds client data, case records, workflows, and client-specific agents.

Product environments

Sable Harbor Professional Plane

Location not recorded

Holds the firm’s protected methods, internal tools, and cross-engagement research.

BS&T railway and facilities

Grouped membership; no reporting line or legal ownership is asserted.

Railway



Blood, Sweat & Tears Railway Company

Taylor, Wyoming

Hauls industrial freight by rail between Wamsutter and Taylor.

Routes

BS&T main line

Wamsutter–Taylor, Wyoming

Carries freight between Wamsutter interchange and Taylor’s industrial facilities.

Routes

East Materials branch

Sweetwater County, Wyoming

Connects the main line to East Materials rail customers.

Routes

Mineral Transfer branch

Sweetwater County, Wyoming

Connects the main line to mineral-loading facilities.

BS&T railway and facilities

Grouped membership; no reporting line or legal ownership is asserted.

Operating facilities

Wamsutter interchange

Wamsutter, Wyoming

Transfers railcars between BS&T and the connecting railroad.

Operating facilities

Taylor Yard and Shops

Taylor, Wyoming

Sorts freight trains and houses locomotive maintenance, fueling, dispatch, and crews.

Operating facilities

East Materials team track

Sweetwater County, Wyoming

Loads and unloads pipe, scrap, and aggregates for rail customers.

Operating facilities

Mineral Transfer team track

Sweetwater County, Wyoming

Loads and unloads minerals between railcars and road transport.

Alexandria systems and publications

System and publication relationships; these cards are not legal entities or reporting departments.

Institutional records and access

Alexandria

Hosting location not recorded

Connects company evidence, decisions, financial records, and historical research.

Institutional records and access

Pinakes

Hosting location not recorded

Provides a searchable catalog and entry point to company records.

Institutional records and access

Canon

Hosting location not recorded

Maintains current company findings and the evidence and changes behind them.

Institutional records and access

Semaphore

Hosting location not recorded

Records and distributes intelligence reports, decision context, corrections, and follow-up.

Alexandria systems and publications

System and publication relationships; these cards are not legal entities or reporting departments.

Institutional records and access

Enterprise Intelligence Brief

Location not recorded

Summarizes current findings, changed views, disagreements, and unanswered company questions.

Personal research tools

Daedalus

Hosting location not recorded

Helps employees search, compare, trace, and analyze authorized company information.

Personal research tools

Daedalus Personal Workspace

Hosting location not recorded

Stores each employee’s private research, comparisons, and working material.

Board committees

Board committee membership and Internal Audit functional oversight.

Board of Directors

Location not recorded

Oversees management, approves major commitments, and appoints senior leadership.

Audit & Compliance Committee

Location not recorded

Oversees financial reporting, controls, compliance, whistleblowing, and independent audit.

Finance & Investment Committee

Location not recorded

Reviews capital allocation, liquidity, borrowing, major investments, and acquisitions.

Technology & Operations Committee

Location not recorded

Oversees technology risk, cybersecurity, operational reliability, and asset integrity.

Board committees

Board committee membership and Internal Audit functional oversight.

Compensation & Human Capital Committee

Location not recorded

Oversees executive pay, succession, leadership development, and workforce policy.

Governance & Nominating Committee

Location not recorded

Reviews Board membership, independence, conflicts, committee structure, and director nominations.

Internal Audit

Location not recorded

Independently audits controls, operations, and governance and reports findings to the Board committee.

Contact collection disciplines

Collection disciplines available to temporary mixed teams; not permanent departmental reporting lines.

Contact

Location not recorded

Collects records, interviews, field observations, and external research for company investigations.

Systems and Digital Collection

Location not recorded

Retrieves authorized records, logs, and data from company systems.

Human Collection

Location not recorded

Interviews people and gathers requested observations.

Field and Technical Collection

Location not recorded

Inspects equipment, facilities, processes, and operating conditions.

Contact collection disciplines

Collection disciplines available to temporary mixed teams; not permanent departmental reporting lines.

JAG Collection

Location not recorded

Collects field-team observations, interviews, artifacts, and operating context.

External Collection

Location not recorded

Researches public, commercial, regulatory, market, and expert sources.

Exploitation

Location not recorded

Collects evidence promptly after incidents, failures, bids, and operating changes.

Directed Observation and Evidence Creation

Location not recorded

Runs measurements, controlled tests, and structured observation when evidence is missing.

J2 Education programs

J2 Education program families, not separate schools or departments.

J2 Education

Location not recorded

Trains employees and executives and operates courses, exercises, and conferences.

Sable Harbor Formation

Location not recorded

Teaches employees how the company, businesses, and shared functions operate.

Executive Formation

Location not recorded

Prepares new senior leaders for company responsibilities and operating decisions.

J2 Professional Education

Location not recorded

Trains intelligence staff through courses, apprenticeships, and continuing education.

Enterprise Professional Programs

Location not recorded

Teaches technical subjects, evidence analysis, decision-making, and business operations.

Convenings and Exercises

Location not recorded

Runs conferences, simulations, case sessions, and company problem-solving events.

Enterprise support services

Administrative membership; Corporate Secretary is a constituent function of OGC.

Enterprise Support Services

Location not recorded

Coordinates shared administration, staffing support, facilities, and service budgets.

Office of the General Counsel

Location not recorded

Provides legal advice and manages corporate legal affairs.

Risk & Compliance

Location not recorded

Tracks enterprise risks, coordinates controls, and manages compliance requirements.

Finance

Location not recorded

Manages financial reporting, treasury, budgets, and capital analysis.

People & Culture

Location not recorded

Recruits employees and manages pay, benefits, employee relations, and career development.

Enterprise Technology Services

Location not recorded

Runs employee technology, shared platforms, infrastructure, and internal automation.

Enterprise support services

Administrative membership; Corporate Secretary is a constituent function of OGC.

Enterprise Security

Location not recorded

Protects company systems and data and manages cybersecurity incidents.

Procurement & Vendor Support

Location not recorded

Manages purchasing, vendors, and supply-chain support.

Safety & Environmental Governance

Location not recorded

Sets company safety and environmental requirements and oversees major exposures.

Quality & Technical Standards

Location not recorded

Maintains shared quality requirements and technical standards.

Facilities & Workplace Services

Location not recorded

Maintains offices, workspaces, and shared facilities.

Corporate Secretary

Location not recorded

Organizes Board meetings, maintains corporate records, and coordinates crisis governance.

Corporate headquarters

Corporate institutions, executive oversight, administrative reporting, and independent audit oversight.

Corporate Headquarters

Sacramento, California

Allocates capital and provides finance, legal, technology, and employee services.

Board of Directors

Location not recorded

Oversees management, approves major commitments, and appoints senior leadership.

Office of the CEO

Location not recorded

Sets company strategy, allocates capital, and leads the businesses.

Enterprise Support Services

Location not recorded

Coordinates shared administration, staffing support, facilities, and service budgets.

Corporate headquarters

Corporate institutions, executive oversight, administrative reporting, and independent audit oversight.



J2

Location not recorded

Collects evidence, investigates company problems, briefs leaders, and trains employees.

Audit & Compliance Committee

Location not recorded

Oversees financial reporting, controls, compliance, whistleblowing, and independent audit.

Internal Audit

Location not recorded

Independently audits controls, operations, and governance and reports findings to the Board committee.

J2 organization

J2 institutional structure; Judgment Watch belongs to Judgment.



J2

Location not recorded

Collects evidence, investigates company problems, briefs leaders, and trains employees.

J2 Headquarters

Location not recorded

Coordinates intelligence work, professional standards, staffing, and support.

Contact

Location not recorded

Collects records, interviews, field observations, and external research for company investigations.

Judgment

Location not recorded

Investigates company problems, tests explanations, and produces evidence-based findings.

J2 organization

J2 institutional structure; Judgment Watch belongs to Judgment.

Orientation

Location not recorded

Briefs leaders on findings, assumptions, risks, and unanswered questions.

Junction Advisory Group

Operating sites

Sends teams into businesses to investigate problems, test improvements, and transfer lessons.

J2 Education

Location not recorded

Trains employees and executives and operates courses, exercises, and conferences.

Judgment Watch

Location not recorded

Screens incoming evidence, connects related signals, and routes material to investigators.

Pinakes portals

Nine catalog entry points into records; same-named functions are not duplicated organizational units.

Pinakes

Hosting location not recorded

Provides a searchable catalog and entry point to company records.

EIB Archive

Hosting location not recorded

Stores issued enterprise intelligence briefs and their historical links.

Finance

Hosting location not recorded

Provides access to financial records and analysis.

Judgment

Hosting location not recorded

Provides live investigations, closed cases, findings, and supporting context.

JAG Knowledge

Hosting location not recorded

Stores field deployments, tested improvements, follow-up, and lessons.

Pinakes portals

Nine catalog entry points into records; same-named functions are not duplicated organizational units.

Semaphore

Hosting location not recorded

Provides intelligence reports and their linked discussions and corrections.

Canon

Hosting location not recorded

Provides current company findings and their earlier versions.

Business Lines

Hosting location not recorded

Connects users to business, product, and operating records.

Collection

Hosting location not recorded

Provides authorized source material and collection records.

Education

Hosting location not recorded

Provides courses, case studies, role learning, and training records.

Enterprise technology capabilities

Enterprise Technology Services product families, not additional departments.

Enterprise Technology Services

Location not recorded

Runs employee technology, shared platforms, infrastructure, and internal automation.

Employee Technology and Experience

Location not recorded

Provides employee devices, workplace software, and technology support.

Enterprise Platforms

Location not recorded

Builds and operates shared company software platforms.

Developer Platform and Engineering Systems

Location not recorded

Provides shared tools and infrastructure for software development.

Enterprise technology capabilities

Enterprise Technology Services product families, not additional departments.

Cloud and Compute Infrastructure

Location not recorded

Operates shared cloud services and computing resources.

Enterprise AI and Automation

Location not recorded

Builds shared AI infrastructure and internal automation.

Technology Architecture and Supply Chain

Location not recorded

Maintains shared technology architecture and technology vendor choices.

Reliability and Service Engineering

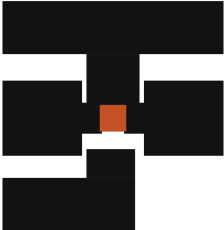
Location not recorded

Keeps shared technology services available and restores them after incidents.

Foundry and Foundry Field

Grouped membership; no reporting line or legal ownership is asserted.

Shared platform



Foundry
Location not recorded

Links operating records across systems while preserving sources, definitions, dates, and changes.

Commercial product



Foundry Field
Location not recorded

Software for production tracking, maintenance, reconciliations, and operating exceptions.

Applications

Operations
Location not recorded

Tracks production, haulage, plan variances, and current operating conditions.

Applications

Maintenance
Location not recorded

Connects equipment, work orders, downtime, and production effects.

Applications

Reconciliation
Location not recorded

Reconciles mine, plant, laboratory, inventory, and financial records.

Applications

Exceptions
Location not recorded

Flags conflicting records and definitions for human review.

Pale Sun and Red Wash

Grouped membership; no reporting line or legal ownership is asserted.

Companies



Pale Sun Inc.

Location not recorded

Develops uranium operations and owns Red Wash Mining.

Companies



Red Wash Mining, LLC

Sweetwater County, Wyoming

Operates the Red Wash uranium mine.

Mine and supporting assets

Red Wash Mine

Sweetwater County, Wyoming

Underground uranium mine and processing plant.

Mine and supporting assets

Red Wash receiving and reagent storage

Red Wash Mine, Wyoming

Receives truck deliveries and stores reagents, steel, and maintenance supplies.

Mine and supporting assets

Red Wash access road

Sweetwater County, Wyoming

Connects Red Wash Mine to the industrial road network.

American Resource Utility Leadership

Named role membership only; no person-to-person reporting relationship is inferred.

Nora Ashcombe President & Chief Operating Officer Joined 1998	Gareth Pike Director of Industrial Operations Joined 2001	Tessa Rourke Controller Joined 2011
Owen Halberg Director of Safety & Environment Joined 2009	Leah Denning People & Administration Manager Joined 2014	Miles Harker Procurement & Systems Manager Joined 2016

Years identify service with the original employer; group acquisition was in 2026.

Named role membership only; no person-to-person reporting relationship is inferred.

Joined 2001

Joined 2004

Joined 2006

Joined 2010

Atlas Meridian Product Leadership

Named role membership only; no person-to-person reporting relationship is inferred.

Simone Vale

Atlas Meridian Product Lead

Year not recorded

Joining years refer to the company. Unrecorded years remain explicit.

Board Committees

Membership of the five standing Board committees; each group identifies its documented Chair.

Audit & Compliance

Thomas “Tom” Calder

Independent Director

Joined 2022

Dr. Lilia Maratovna Galiullina

Independent Director

Joined 2020

Margaret “Maggie” Kincaid

Independent Chair

Joined 2023

Chair: Thomas “Tom” Calder

Years identify entry to the Board, not appointment to this committee. Part 1 of 5.

Board Committees

Membership of the five standing Board committees; each group identifies its documented Chair.

Finance & Investment

Hendrik “Harry” van der Velde

Independent Director

Joined 2019

Marianne Voss

Director

Joined 2022

Evan Mercer

Director

Joined 2021

Daniel Mercer

Director

Joined 2016

Chair: Hendrik “Harry” van der Velde

Years identify entry to the Board, not appointment to this committee. Part 2 of 5.

SH-ORG-PEOPLE-BOARD-COMMITTEES

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Board Committees

Membership of the five standing Board committees; each group identifies its documented Chair.

Technology & Operations

Dr. Lilia Maratovna Galiullina
Independent Director

Joined 2020

Priya Raman
Director

Joined 2016

Margaret “Maggie” Kincaid
Independent Chair

Joined 2023

Evan Mercer
Director

Joined 2021

Chair: Dr. Lilia Maratovna Galiullina

Years identify entry to the Board, not appointment to this committee. Part 3 of 5.

Board Committees

Membership of the five standing Board committees; each group identifies its documented Chair.

Compensation & Human Capital

Margaret “Maggie” Kincaid

Independent Chair

Joined 2023

Dr. Lilia Maratovna Galiullina

Independent Director

Joined 2020

Hendrik “Harry” van der Velde

Independent Director

Joined 2019

Chair: Margaret “Maggie” Kincaid

Years identify entry to the Board, not appointment to this committee. Part 4 of 5.

Board Committees

Membership of the five standing Board committees; each group identifies its documented Chair.

Governance & Nominating

Margaret “Maggie” Kincaid

Independent Chair

Joined 2023

Thomas “Tom” Calder

Independent Director

Joined 2022

Jon Bell

Non-management Director

Joined 2016

Chair: Margaret “Maggie” Kincaid

Years identify entry to the Board, not appointment to this committee. Part 5 of 5.

Named role membership only; no person-to-person reporting relationship is inferred.

Joined 1992

Joined 2008

Joined 1999

Joined 2007

Joined 2010

Project Cradle Team

Named role membership only; no person-to-person reporting relationship is inferred.

Kenji Arakawa Head of Cradle Year not recorded	Tessa Quinn Economic Geologist Year not recorded	Luis Ortega Process Engineer Year not recorded
Maeve Donnelly Data Engineer Year not recorded		

Joining years refer to the company. Unrecorded years remain explicit.

Enterprise Leadership

Named role membership only; no person-to-person reporting relationship is inferred.

Daniel Mercer Chief Executive Officer Joined 2016	Priya Raman Head of Product Joined 2016	Elena Torres Head of Customer Delivery Year not recorded
Marcus Reed Principal Engineer Year not recorded	Dr. Maya Okafor Scientific Reviewer Year not recorded	Caleb Hargrove Head of Field Operations Year not recorded
Rachel Sloane Director of Advanced Programs Year not recorded		

Joining years refer to the company. Unrecorded years remain explicit.

Foundry and Customer Delivery

Named role membership only; no person-to-person reporting relationship is inferred.

Priya Raman Head of Product Joined 2016	Elena Torres Head of Customer Delivery Year not recorded	Marcus Reed Principal Engineer Year not recorded
Dr. Maya Okafor Scientific Reviewer Year not recorded	Caleb Hargrove Head of Field Operations Year not recorded	Nadia Software Engineer Year not recorded

Joining years refer to the company. Unrecorded years remain explicit.

Pale Sun and Red Wash Leadership

Named role membership only; no person-to-person reporting relationship is inferred.

Evan Vilander President Joined 2021	Mari Varela Chief Operating Officer, Pale Sun; Chief Executive Officer, Red Wash Year not recorded	Cole Site Superintendent Year not recorded
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Joining years refer to the company. Unrecorded years remain explicit.

Willow and Advanced Programs

Named role membership only; no person-to-person reporting relationship is inferred.

Gid Voss Head of Willow Joined 2018	Rachel Sloane Director of Advanced Programs Year not recorded
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Joining years refer to the company. Unrecorded years remain explicit.

Willow Research Team

Named role membership only; no person-to-person reporting relationship is inferred.

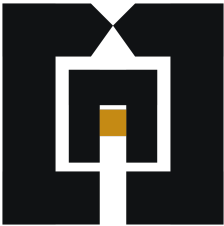
Gid Voss Head of Willow Joined 2018	Mara Aquil Embedded Systems Engineer Year not recorded	Theo Bell Applied Mathematician Year not recorded
Benji Rao Mechanical Engineer Year not recorded	Jun Park Interaction Designer Year not recorded	Ellis “Eli” Hoberg RF & Communications Engineer Year not recorded
Owen Kessler Junior Research Engineer Joined 2025	Layla Haddad Research Engineer Joined 2025	

Joining years refer to the company. Unrecorded years remain explicit.

Project Cradle

Grouped membership; no reporting line or legal ownership is asserted.

Business



Project Cradle

Fairmont area, West Virginia; Tasmania deployments

Recovers rare-earth materials from industrial byproducts and mine water for sale to specialist refiners.

Development and upgrading

Bedford

Fairmont area, West Virginia

Builds recovery units and tests, blends, and upgrades recovered rare-earth material.

Recovery installations

Stream 17 recovery installation

Tasmania, Australia

Recovers mixed rare-earth concentrate from Kelly Gang Mining’s mineral-processing side stream.

Recovery installations

Demotte Gen 1 recovery installation

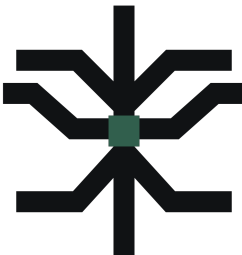
North-central West Virginia

Recovers rare-earth material from a controlled mine-water slipstream.

Willow and the Fort

Grouped membership; no reporting line or legal ownership is asserted.

Research program



Willow

Pittsburgh area, Pennsylvania

Builds and tests industrial prototypes, sensors, software, and experimental processes.

Research site

The Fort

Pittsburgh area, Pennsylvania

Houses Willow’s laboratories, prototype workshops, equipment storage, and visiting staff.

Buildings and grounds

The Big Shed

The Fort, Pittsburgh area, Pennsylvania

Houses research laboratories, prototype construction, instrumentation, and testing.

Buildings and grounds

The Small Shed

The Fort, Pittsburgh area, Pennsylvania

Houses administrative offices and temporary accommodation for visiting staff.

Buildings and grounds

The White Shed

The Fort, Pittsburgh area, Pennsylvania

Receives, quarantines, stages, and stores equipment, samples, and field supplies.

Buildings and grounds

The Museum

The Fort, Pittsburgh area, Pennsylvania

Working yard holding retired prototypes, industrial equipment, and outdoor test materials.

Project Cradle — external hosts

Grouped membership; no reporting line or legal ownership is asserted.

External host operators



Kelly Gang Mining
Tasmania, Australia

Mines and processes industrial and heavy minerals.

External host operators



Demotte Reclamation Services
North-central West Virginia

Treats drainage water from a former underground coal complex.

External investors

Grouped membership; no reporting line or legal ownership is asserted.

Outside capital providers

Harrison Vale Partners

Chicago, Illinois

Invests growth capital in enterprise, data, and industrial software businesses.

Outside capital providers

Wolf Ridge Holdings

Chicago / Midwest

Invests long-term capital in infrastructure,
resources, manufacturing, and industrial technology.

Historical opportunities

Grouped membership; no reporting line or legal ownership is asserted.

Opportunities considered

Cedar Junction Conversion Services, LLC

Kentucky

Owned an idled downstream processing property considered for acquisition.

Opportunities considered

Juniper Mesa

Western New Mexico

Uranium-property opportunity considered for acquisition.

Opportunities considered

Mesa Lantern Minerals, LLC

Location not recorded

Offered the Juniper Mesa uranium-property rights for sale.

Opportunities considered

Salt Rim Recovery Venture

Western United States; site not recorded

Proposed restarting a uranium-recovery operation.

The Original Eight

Historical founding-cohort membership, not current employee headcount or reporting authority.

Daniel Mercer Chief Executive Officer Joined 2016	Priya Raman Head of Product Joined 2016	Jon Bell Non-management Director Joined 2016
Elena Torres Head of Customer Delivery Year not recorded	Marcus Reed Principal Engineer Year not recorded	Dr. Maya Okafor Scientific Reviewer Year not recorded
Caleb Hargrove Head of Field Operations Year not recorded	Rachel Kim Finance & Operations Lead Year not recorded	

Titles show latest recorded roles. Rachel Kim left in 2024; Jon Bell left management in 2023.

Research artifacts

Grouped membership; no reporting line or legal ownership is asserted.

Research models and prototypes

Ned Kelly Mark I

Location not recorded

Screened deposits and residues for rare-earth recovery prospects.

Research models and prototypes

Ned Kelly Mark II

Location not recorded

Finds industrial processes that concentrate recoverable rare-earth material.

Research models and prototypes

Theseus

Klein / Willow, Pittsburgh area, Pennsylvania

Maps radio propagation underground.

Research models and prototypes

Theseus II

Klein / Willow, Pittsburgh area, Pennsylvania

Maps radio propagation underground.

Research artifacts

Grouped membership; no reporting line or legal ownership is asserted.

Research models and prototypes

Theseus III

Klein / Willow, Pittsburgh area, Pennsylvania

Maps radio propagation underground.

Research models and prototypes

Mara’s field data logger

Location not recorded

Records field sensor measurements.

Research models and prototypes

Benji’s vibration test rig

Location not recorded

Tests equipment under vibration.

Research history

Grouped membership; no reporting line or legal ownership is asserted.

Historical operating programs and site



Klein
Pittsburgh area, Pennsylvania

Developed industrial prototypes and performed coal-sector engineering work.

Historical operating programs and site

Klein historical shop
Pittsburgh area, Pennsylvania

Housed early industrial engineering, machine tools, and prototype testing.

Historical operating programs and site



Project Emberline
Charleston, West Virginia

Delivered coal-sector customer work, field trials, and paid industrial experiments.

Projects and research models

Project Janus
Klein shop, Pittsburgh area, Pennsylvania

Tested reconstruction of operating events from conflicting records.

Projects and research models

Project Glasshouse
Test site not recorded

Tested underground machine vision.

Projects and research models

Project Mule
Willow; field sites not recorded

Built rugged sensor rigs for field use.

Research history

Grouped membership; no reporting line or legal ownership is asserted.

Projects and research models

Project Lantern

Customer site not recorded

Investigated recurring production losses across operating and maintenance records.

Projects and research models

Wallaby

Location not recorded

Evaluated rare-earth recovery from old tailings and process residues.

Projects and research models

Atlas research lineage

Location not recorded

Explored software for investigations across industrial records.

Projects and research models

Hound

Willow, Pittsburgh area, Pennsylvania

Experimental agent that searched tools and records during industrial investigations.

Projects and research models

Ranger

Willow, Pittsburgh area, Pennsylvania

Tested agent permissions, investigation rules, and evidence requirements.

Business origins and former counterparties

Grouped membership; no reporting line or legal ownership is asserted.

Former Red Wash seller



Northstar Minerals, Inc.

Location not recorded

Former owner and seller of Red Wash Mining.

External operating and founder history



Quality Forest Communications

Location not recorded

Builds radio relay and repeater networks for forestry and mountain operations.

External operating and founder history

Argent Ridge Mining

Nevada operations; office not recorded

Operated the Blackridge copper-and-gold mine.

External operating and founder history

Blackridge Mine

Nevada

Open-pit copper-and-gold mine operated by Argent Ridge Mining.

External operating and founder history

Gid Voss’s balloon business

Location not recorded

Provides hot-air-balloon flights.

Railway predecessor

Bloodstone & Southern Railway

Wyoming

Carried freight before BS&T’s 1954 reorganization.